Internet infrastructure

Architecturally speaking, most of the conventional models of network and system monitoring are (semi) centralised and can suffer from a single point of failure.

Measuring the characteristics of network traffic is critical to identify and diagnose performance problems. Some of the metrics are traffic volume, delay, delay distribution, packet loss rate, etc. The insight we gain is useful for designing distributed services.

The Internet is one of the most important distributed network infrastructures today. Interestingly, performance was not a design goal at the time of its conception.

Most network measurement models utilise a monitoring approach that is either 'probe based' (active) or 'router based' (passive).

P2P network monitoring

The static design and rigidity of Internet architecture makes it difficult to innovate or deploy technologies in the core.

Consequently, most of the research and innovation has focused on 'overlay' and 'peer-to-peer' systems. A data structure like distributed hash tables (DHT) is used to implement protocols that keep track of the changes in membership and thus the attendant changes in the route. Tapestry, Chord, Kelips and Kademlia are the common implementation protocols.

In order to monitor such topology, lookup performance has often been measured with hop count, latency, success rate and probability of timeouts. For applications, lookup latency is used as a unified performance metric.

Cloud and container infrastructures

The rapid development of sensor technologies and wireless sensor network infrastructure has led to the generation of huge volumes of real-time data traffic. Cloud infrastructure needs to elastically handle the data surge, and provide computing and storage resources to applications in real-time.

As a result, software infrastructure that supports analytics for real-time decision support must have the ability to enable traffic data acquisition, and distributed and parallel data management. Intrusive network monitoring practices will neither work nor scale in dense cloud infrastructures. Cloud lifecycle management integrated with measurement is a value proposition.

Networked storage infrastructure

Storage networking infrastructures like SAN (storage area networks), NAS (network attached storage), NAS filers, VTL (virtual tape libraries), pNFS (parallel Network File System), DRBD (distributed replicated block device) clusters and RDMA (remote direct memory access) zero-copy networking have very stringent network performance requirements supported by backplane fabric technologies like data centre Ethernet, fibre channel (FC) and Infiniband. The common theme is very high throughput and very low latency.

Advanced Telecommunications Computing Architecture (ATCA)

Mobile data will grow ten times each year for the rest of this decade. A typical ATCA platform offers a mix of CPU, DSP, storage and I/O payload blades, together with up to four switch blades. The network and server infrastructure for mobile and fixed line services requires many line rate functions.

The functions include security, deep packet inspection (DPI), packet classification and load balancing. Line rate DPI enables advanced policy management and gives service providers full control of network resources. As network capacity increases, the line rates must increase from 10 Gbps and 40 Gbps to hundreds of Gbps, in the future.

Existing approaches of probe based network monitoring will not work and require the monitoring models to deeply integrate with line rate functions. Techniques like Packet Doppler represent a promising solution as they monitor SLA metrics.

Analytics based approaches

The current data-driven approach to defining metrics has led to operation analytics and route analytics, among other techniques, being used to manage and monitor heavy bandwidth overlay networks.

While topology discovery products are established, there are also many logical network 'overlays' for the topology, including network routes. As a result, device-centric discovery mechanisms cannot capture the routes through the network.

Route analytics solves the configuration management requirements of organisations with highly meshed networks. By capturing the complex network topology relationships, route analytics systems can perform advanced configuration management, incident correlation and root-cause identification.

Performance metrics like latency, throughput and cost can be calculated since the analytics systems participate in the cooperative negotiations between routers. Real-time alerts can be generated as route paths become modified or there is a problem related to routing protocol. This can trigger polling, which leads to root-cause analysis through the event correlation system.

Packet Doppler

Delay and loss sensitive applications such as telecommuting, streaming media, gaming and live conference applications require stringent monitoring of service level agreement (SLA) metrics.

The traditional methods of end-to-end monitoring are not effective, as they are based on active probing and thus inject measurement traffic into the network.